

Chemistry, 2014 EC, natural science, entrance exam

Grade 12 · Chemistry · 2014 E.C.

Time allowed: 120 minutes

Questions: 30

Total marks: 30

Name _____ Grade _____ ID No. _____ Date _____

1. What is the purpose of synthetic Cryolite in the extraction of aluminum by the Hall's process? (1)
 - A. Cryolite reduces the melting point of alumina
 - B. Cryolite increases the purity of the extraction of aluminum.
 - C. Cryolite is used as a catalyst in the extraction of aluminum.
 - D. Cryolite increases the melting point of alumina
2. Which of the following is a chemical property of aluminum? (1)
 - A. Aluminum is a grey metal in color.
 - B. Aluminum liberates hydrogen gas from dilute acids like HCl and H₂SO₄
 - C. Aluminum has a lower density than any other commercial metal.
 - D. Aluminum is a good conductor of heat and electricity.
3. How is tanning carried out? In tanning the hides or skins are (1)
 - A. soaked in lime and water for several days.
 - B. soaked in water for several days.
 - C. treated with dilute hydrochloric acid to remove the line.
 - D. treated with chromium salts.
4. How can a project of establishing an industry that consumes coal affect the environment? (1)
 - A. Rise in global temperature
 - B. Increment in concentration of nitric oxide
 - C. By changing the PH of water
 - D. Production of high level of lead in the atmosphere
5. Which of the following is a fundamental quantity SI unit? (1)

A. Force	B. Speed
C. Length	D. Pressure
6. What is enthalpy change? It is (1)
 - A. the amount of heat given out or absorbed in a reaction at constant pressure.
 - B. the amount of heat given out in a reaction.
 - C. the amount of heat absorbed in a reaction.
 - D. the amount of heat given out or absorbed in a reaction at constant pressure and volume.
7. What is oxidation? Oxidation is (1)
 - A. the loss Of electrons by a species, accompanied by an increase in oxidation number.
 - B. the addition of hydrogen to a substance and the removal of oxygen from a substance.
 - C. the reduction of the oxidation state of the oxidizing agent.
 - D. the addition of electron (s) to the oxidizing agent.

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8. What is meant by preferential discharge, if a number in a solution? It means that last at given electrode. (1)
- A. least energy will be liberated first a given electrode.
 - B. least energy will be liberated last at a given electrode.
 - C. highest energy will be liberated first at a given electrode.
 - D. highest energy will be liberated fist at
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9. Faraday' s Second Law states: (1)
- A. passing the same quantity of electricity through solutions of different electrolytes produces masses of substances that are directly proportional to their equivalent masses.
 - B. passing the same quantity of electricity through solutions of different electrolytes produces masses of substances that are directly proportional to their molecular masses.
 - C. the amount of substance consumed or produced at an electrode is inversely proportional to the quantity of electricity passing through the solution.
 - D. the amount of a substance consumed or produced at an electrode is directly proportional to the quantity of electricity passing through the solution.
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10. Which of the following is true about Zn - Cu voltaic cell? (1)
- A. After the cell runs for several hours, the Zn electrode weights less.
 - B. The anode half - cell consists of Cu electrode dipped into a $\text{Cu}^{(2+)}$ solution.
 - C. As the cell operates, electrons are continuously generated at the cathode and consumed at the anode.
 - D. The electrons flow from right to left through the external circuit.
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11. Which of the following postulates of Dalton' s atomic theory was modified by the modern atomic theory? (1)
- A. Atoms are indivisible and indestructible.
 - B. Atoms are the smallest particles of all elements.
 - C. Compounds are formed when atoms of more than one element combine.
 - D. Atoms are neither created nor destroyed in chemical reactions.
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12. Which of the following includes all of the known atomic models? (1)
- A. Bohr Model, the Quantum Mechanical Model, Thomson Model, Rutherford Model
 - B. Bohr Model, Quantum Mechanical Model, Dalton's Atomic Model, Valence Bond Model
 - C. Bohr Model, Thomson Model, Rutherford Model, Dalton's Atomic Model, de Broglie Model
 - D. Bohr Model, the Quantum Mechanical Model, Rutherford Model, Thomson Model, Dalton's Atomic Model
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13. What does Mendeleev' s Periodic Law state? Properties of elements are periodic functions of their (1)
- | | |
|-------------------|----------------------------|
| A. atomic masses | B. atomic radius |
| C. atomic numbers | D. electron configurations |
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14. Which of the following statements is correct regarding alkenes and alkynes? (1)
- A. Alkenes are soluble in polar solvents, while alkynes are not soluble in polar solvents.
 - B. Alkenes and alkynes are soluble in polar solvents.
 - C. The molecule of alkenes and alkynes are held together by weak inter-molecular force.
 - D. Alkynes have longer bond length and less surface area than alkenes.
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15. Which functional group containing compound is named using the suffix "-ol"? (1)
- | | |
|-----------|-------------|
| A. Aryl | B. Carboxyl |
| C. Phenyl | D. Hydroxyl |

16. What is a Lewis acid? It is _____ (1)

- A. an electron pair acceptor
B. a proton acceptor
C. a proton donor
D. an electron pair donor

17. Which Of the following statements is NOT correct about minerals and ores? (1)

- A. All ores are minerals
- B. All minerals are ores
- C. Ores are usually used to extract metals economically
- D. Minerals are the natural forms in which various metals exist.

18. Which of the following is NOT a major step in metallurgical processes? (1)

- A. Concentrating the ore
B. Mining the ore
C. Extracting the metal and refining
D. Pretreatment

19. Which monomers are used for the synthesis of Teflon? (1)

- A. Propylene
B. Tetrafluoroethylene
C. Isoprene
D. Styrene and Phenol

20. What is the significance of electron probability distribution? It tells us about (1)

- A. finding the electron in its orbit at a particular time.
B. the uncertainty to assign fixed path for electrons.
C. the probability of finding an electron is very far away from the nucleus.
D. finding the electron very close to the nucleus at all times.

21. Which Of the following is a property Of covalent compounds? Covalent compound (1)

- A. has high melting and boiling points
- B. is non-volatile
- C. is crystalline solids at room temperature
- D. is generally insoluble in water

22. Which of the following is a covalent molecule? (1)

- A. NaCl
B. PCl_3
C. $(\text{CO}_3)^{2-}$
D. $[\text{NH}_4]^+$

23. Which Of the following carboxylic acids is a solid substance at room temperature? (1)

- A. Octanoic acid
B. Decanoic acid
C. Heptanoic acid
D. Nonanoic acid

24. What is the IUPAC name of an ester having the molecular formula $C_6H_{12}O_2$ (1)

- A. Methyl butyrate
B. Pentyl butyrate
C. Ethyl butanoate
D. Ethyl butyrate

25. What is the rate of dissolution? It is (1)

- A. the speed at which a solid substrate goes into solution.
- B. the rate at which a solid turns to liquid
- C. the speed at which a solid sublimates.
- D. the rate at which a solid dissociates.

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- 26.** Which one of the following is correctly stated regarding solubility as an equilibrium process? (1)
- A. Saturated solution is a solution in which more of the solute can be dissolved.
 - B. Unsaturated solution is a solution to a given dissolved substance.
 - C. Supersaturated solution is more stable than saturated solution.
 - D. Supersaturated solution is a solution that contains more than the equilibrium amount of dissolved solute.
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- 27.** Which of the following assumptions of the kinetic molecular theory of gases explains the high compressibility of gases? (1)
- A. There is no force of attraction between gas molecules.
 - B. The actual volume of gas molecules is negligible.
 - C. Different particles have different speeds.
 - D. Particles are always in random motion.
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- 28.** Consider a bottle containing water, water vapor and air in equilibrium. If the temperature decreases slightly, the amount of water vapors; (1)
- A. will increase and the vapor pressure will decrease.
 - B. and vapor pressure will increase.
 - C. and vapor pressure will decrease.
 - D. will decrease and the vapor pressure will increase.
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- 29.** Why do water droplets collect on the outside of a glass of water on a hot day? (1)
- A. Water from the glass starts to vaporize.
 - B. There is heat exchange between the water in the glass and the surrounding.
 - C. There will be a sublimation of water vapor on the outside of a glass.
 - D. There will be a condensation of water vapor on the glass.
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- 30.** What are hydrocarbons? Compounds that contain (1)
- A. hydrogen and carbon only.
 - B. hydrogen and carbon with no benzene rings.
 - C. benzene and its derivatives only.
 - D. benzene rings only

— END OF EXAMINATION —